



UKAEA

FISPACT-II

Solver tolerances

Introduction

FISPACT

* Tolerance example

<< ----- set initial conditions - >>

MASS 0.80 12

Be 62.6239

Al 36.9566

O 0.1930

C 0.0965

Mn 0.0386

Cr 0.0289

Ta 0.0289

Au 0.0096

W 0.0096

Mo 0.0096

Hg 0.0029

Co 0.0019

MIND 1.0E+05

HALF

DOMINANT NUCLEIDES

NUCLIDE	ACTIVITY	PERCENT	NUCLIDE	HEAT	PERCENT	NUCLIDE	DOSE RATE	PERCENT
	(Bq)	ACTIVITY		(kW)	HEAT		(Sv/hr)	DOSE RATE
Total	1.4688E+08		Total	1.9520E-08		Total	2.2806E-02	
1 Au198	1.1197E+08	76.23E+00	Au198	1.3080E-08	67.01E+00	Au198	1.2029E-02	52.74E+00
2 Ta182	2.0563E+07	14.00E+00	Ta182	4.9407E-09	25.31E+00	Ta182	9.0526E-03	39.69E+00
3 W 187	9.5297E+06	64.88E-01	W 187	1.1350E-09	58.15E-01	W 187	1.1727E-03	51.42E-01
4 Cr 51	2.4693E+06	16.81E-01	Mn 56	2.1584E-10	11.06E-01	Mn 56	3.4521E-04	15.14E-01
5 Hg197	7.9172E+05	53.90E-02	Co 60	6.9832E-11	35.78E-02	Co 60	1.5396E-04	67.51E-02
6 Mn 56	5.3218E+05	36.23E-02	Mo 99	2.2975E-11	11.77E-02	Cr 51	1.8655E-05	81.80E-03

Target nuclide Tc101 99.999% of inventory given by 1 path

path 1 99.999% Mo100 --- (R) --- Mo101 --- (d) --- Tc101 --- (S) ---
100.00% (n.g) 100.00% (b-)

Target nuclide Ta182n 99.997% of inventory given by 1 path

Inventory rate equation

- coupled ODE

$$\frac{dN_i}{dt} = -N_i(\lambda_i + \sigma_i\phi) + \sum_{j \neq i} N_j(\lambda_{ji} + \sigma_{ji}\phi)$$

- σ_{ji} energy-dependent reaction cross sections for $j \rightarrow i$ reactions, σ_i sum of all $i \rightarrow j$ reactions

$$\sigma_{ji} = \frac{\int \tilde{\sigma}_{ji}(E)\phi(E)dE}{\int \phi(E)dE} \approx \frac{\sum_k \tilde{\sigma}_{ji}(E_k)\phi_k}{\sum_k \phi_k}$$

- λ_{ji} decay constants for $j \rightarrow i$, λ_i sum of all $i \rightarrow j$ decays
- total flux ϕ (from transport)

Compactly, at the heart of FISPACT-II is the inventory equation:

$$\dot{\mathbf{N}}(t) = \mathbf{A}\mathbf{N}(t) \quad (1)$$

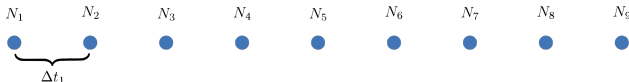
where $\mathbf{A}(\lambda, \sigma, \phi) \in \mathbb{R}^n \times \mathbb{R}^n$, λ, σ, ϕ are decay constants, reaction cross-section and the flux respectively and $\mathbf{N} : \mathbb{R}^+ \rightarrow \mathbb{R}^n$, n is the number of nuclei.

- Currently solved using LSODES (Livermore Solver for Ordinary Differential Equations)

Numerical setting

Within FISPACT-II (LSODE):

$$N(t_i) = N_i$$



- $\dot{N} = \lim_{\Delta t \rightarrow 0} \frac{N_{t+1} - N_t}{\Delta t}$
- Large range of decay constants.
 - slow and rapidly decaying isotopes
 - large range of eigenvalues
 - Require a large 'region of stability' (A-stable)
- Implicit Backwards Difference Formula (BDF):

$$\sum_{k=0}^s a_k N_{i+k} = \Delta t A N_{i+s}, \text{ for the order } s \text{ approximation}$$

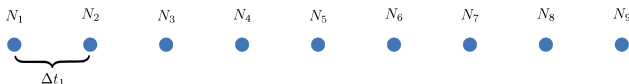
- Solver: LU factorising with pivots

$$P(\mathbb{I}_{n \times n} - \Delta t A) N_{i+s} = L U N_{i+s}$$

Numerical setting

Within FISPACT-II (LSODE):

$$N(t_i) = N_i$$



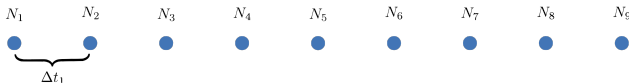
- Finite difference approximations have a local truncation error, measures the error introduced by each step. This is estimated as a Taylor expansion.
- First time step estimated by spectral estimation.
- BDF local truncation error is estimated as:

$$\tau(N_i, s) = \frac{\Delta t_i^{s+1} N_i^{(s+1)}}{1 + s}$$

- The size of the time-step is estimated based on $\tau(N_i, s)$

Within FISPACT-II (LSODE):

$$N(t_i) = N_i$$



- Finite difference approximations also have a global truncation error:
- Absolute error places a constraint (within LSODES) on the maximum rate of change.

FISPACT

• Tolerance example

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HALF

UNCERTAINTY 2

TOLERANCE 0 1.0E2 2.0E-4

TOLERANCE 1 1.0E2 2.0E-3

TOLERANCE *itol atol rtol*

- *itol*: 0 full solver settings, 1 Uncertainty solver settings
- *atol*: Absolute error settings (10^4)
- *rtol*: Relative tolerance (2×10^{-3})

FISPACT

• Tolerance example

```
<< ----- set initial conditions - >>
```

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DOMINANT NUCIDES

NUCLIDE	ACTIVITY	PERCENT	NUCLIDE	HEAT	PERCENT	NUCLIDE	DOSE RATE	PERCENT
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2 Ta182	2.0567E+07	13.95E+00	Ta182	4.9418E-09	25.28E+00	Ta182	9.0545E-03	39.71E+00
3 W 187	9.6362E+06	65.36E-01	W 187	1.1477E-09	58.71E-01	W 187	1.1858E-03	52.01E-01
4 Cr 51	2.4712E+06	16.76E-01	Mn 56	1.6495E-10	84.38E-02	Mn 56	2.6381E-04	11.57E-01
5 Hg197	7.9560E+05	53.96E-02	Co 60	6.9833E-11	35.72E-02	Co 60	1.5397E-04	67.53E-02
6 Mn 56	4.0670E+05	27.58E-02	Mo 99	2.3088E-11	11.81E-02	Cr 51	1.8669E-05	81.88E-03

Target nuclide Tc101 100.000% of inventory given by 1 path

```
path 1 100.000% Mo100 --- (R) --- Mo101 --- (d) --- Tc101 --- (S) ---
100.00%(n,g) 100.00%(b-)
```

Target nuclide Ta182n 99.997% of inventory given by 1 path

• Simple convergence test

	TOL. 0 1.0E4 2.0E-3	TOL. 0 1.0E2 2.0E-6	TOL. 0 1.0E1 2.0E-8
Mn 56 Activation	5.3218E+05	4.0636E+05	4.0621E+05

We learned about:

- Numerical stability
- FISPACT-II can require careful monitoring of tolerance settings
- Tolerance
 - Controls the absolute and local truncation error of the LSODES solver.
- Numerical convergence test
 - The best way to estimate the correct tolerance values is by a convergence test